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RESEARCH

Dynamic graph-guided imputation network for partial multi-view incomplete multi-label classification

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Abstract

In practice, multi-view multi-label classification often faces the dual challenge of missing views and labels. Existing methods typically avoid redundant computations by simply masking missing items, which neither recovers missing view information nor provides effective supervision for unknown labels, leading to underutilized information. To address this, we propose the Dynamic Graph-Guided Imputation Network (DGGI-Net), which centers on high-quality view completion. Specifically, DGGI-Net adaptively constructs intra-view similarity graphs and aggregates available neighborhood features on these graphs to impute missing views, thereby restoring more complete representations and reducing cross-view bias. Concurrently, we design a dual-path fusion strategy that aggregates original and imputed views into two separate branches for classification learning, preserving reliable information while mitigating cross-view contamination. Furthermore, we introduce the Category-Adaptive Pseudo-Label Augmentation (CAPLA) mechanism. Based on the empirical class distribution of known labels, CAPLA sets dual thresholds for each category and discards low-confidence instances, generating high-confidence pseudo-labels for unlabeled instances, alleviating supervision sparsity while keeping the pseudo-label distribution consistent with the empirical class distribution under class imbalance. Experiments on five public datasets demonstrate that our method consistently outperforms strong baseline methods.

Keywords Partial multi-view learning · Incomplete multi-label classification · Pseudo-labeling · View imputation

1 Introduction

Multi-view learning has become a core paradigm in machine learning and computer vision, as it exploits complementary information from heterogeneous sources or feature extractors (Wen et al. 2021; Ding et al. 2025). It has been successfully applied to recommendation, biometrics, and image/video annotation (Kang et al. 2025; Li et al. 2025a). Building on this foundation, multi-view multi-label classification (MVMLC) extends the setting to instances associated with multiple, non-mutually-exclusive labels while still leveraging information from multiple views (Zou et al. 2023; Zhang et al. 2020). This formulation is well suited to complex, multi-faceted data in which label co-occurrence is common (Li et al. 2025b). For example, medical diagnosis may involve complementary imaging modalities such as MRI and CT while a patient may suffer from multiple co-occurring pathologies (Wang et al. 2026); autonomous driving perception often combines camera and LiDAR information to recognize multiple objects and scene attributes. These applications require models to jointly exploit cross-view complementarity and label co-occurrence, which distinguishes MVMLC from conventional single-label classification (Zhao et al. 2023b).

Driven by practical demands to jointly model multiple views and multiple labels, recent MVMLC studies have improved prediction performance from several perspectives, including latent semantic representation, matrix factorization, graph-based label correlation modeling, and deep representation learning. Representative methods such as M3Lcmf (Xing et al. 2019), AC-MIMLLN (Li et al. 2020), GRADIS (Chen et al. 2020), and LCBM (Sun and Zong 2021) enhance cross-view fusion, multi-instance label modeling, or label correlation learning from different perspectives. Although these methods have achieved promising performance, many of them are mainly designed for complete-view and fully

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labeled scenarios. In practice, however, data acquisition and annotation are often costly, noisy, and incomplete, making it difficult to obtain all views and all labels for every sample.

The simultaneous absence of views and labels, namely dual incompleteness, poses a more realistic and challenging setting for MVMLC (Liu et al. 2023a). For instance, in clinical workflows, a patient's CT scan may be unavailable due to storage or protocol constraints, while critical diagnostic labels may remain unannotated because of limited resources. View-level incompleteness compromises cross-modal complementarity, whereas label-level incompleteness weakens supervision and label correlation modeling; together, these issues degrade the reliability of learned representations in practical deployments (Liu et al. 2022, 2021). These realities call for algorithms that are robust to both forms of missingness, and recent years have seen methods tailored for dual-incomplete MVMLC. For example, DICNet (Liu et al. 2023b) proposes a deep contrastive learning framework with stacked autoencoders to capture view-specific features, and introduces an incomplete contrastive loss to align cross-view semantic consistency. MTD (Liu et al. 2023c) separates each view into shared and view-specific features using dual encoders, and combines cross-channel contrastive learning with label-guided graph regularization to handle incomplete multi-view weak multi-label learning. RANK (Liu et al. 2025) adopts a label-driven multi-view contrastive scheme that preserves label-guided structure to avoid false negatives and dynamically reweights views through a quality-aware subnetwork for robust fusion. AIMNet (Liu et al. 2024a) employs a cross-view attention-guided embedding imputer for reconstruction and a GAT-based label semantic fusion with confidence weighting for incomplete multi-view partial multi-label learning. Although existing methods have made some progress, several issues remain insufficiently addressed when missing views and missing labels occur simultaneously. Many current deep learning methods, including DICNet, MTD, and RANK, still largely rely on missing-view indicators and label-observation masks to prevent unavailable views or unobserved labels from directly participating in training. Although such strategies can reduce the interference caused by missing data, they may also prevent models from fully recovering and exploiting the latent complementary information in missing views, thereby weakening the advantages of multi-view learning. Even when some methods incorporate view recovery modules, they typically do not explicitly address the noise contained in the recovered views; such noise may propagate through the unified fusion process and ultimately degrade the quality of feature representations (Wang et al. 2024a, 2025). For missing labels, pseudo-label generation is a commonly used supplementary supervision strategy. However, most existing pseudo-labeling strategies adopt fixed thresholds or global confidence criteria, lacking explicit modeling of class imbalance and label sparsity. Consequently, they struggle to maintain stable pseudo-label quality and coverage across different categories.

To address these challenges, we propose the Dynamic Graph-Guided Imputation Network (DGGI-Net), a unified framework for multi-view multi-label learning with missing views and labels. DGGI-Net consists of three key components. First, it performs graph-guided view imputation by constructing sample-wise similarity graphs and exploiting cross-view neighborhood information to recover missing views. Second, it introduces a dual-path fusion strategy to separate observed and imputed views, allowing reliable observed information to dominate the final prediction while using recovered views to provide supplementary supervision with reduced noise propagation. Third, it develops a Category-Adaptive Pseudo-Label Augmentation (CAPLA) mechanism, which calibrates category-specific thresholds according to global label statistics to generate high-confidence pseudo-labels under label sparsity and class imbalance.

In summary, our paper makes the following key contributions:

- We propose a dynamic graph-guided view imputation module that constructs sample-wise similarity graphs and applies graph convolutions to recover missing views with high fidelity. In addition, we introduce a neighborhood regularization term that prevents geometric distortions induced by graph aggregation.
- We propose a dual-path multi-view fusion strategy that aggregates original and imputed views in separate branches, preserving reliable information while isolating reconstruction noise; a view-consistency contrastive loss is further introduced to enhance the quality of imputed views by promoting cross-view semantic consistency.
- We design a category-adaptive pseudo-label augmentation mechanism that calibrates category-specific confidence thresholds from global label statistics to generate high-confidence pseudo-labels for unlabeled instances, thereby alleviating supervision sparsity.

2 Related work

Given that our study lies at the intersection of multi-label prediction and multi-view representation learning, this section surveys related work with an emphasis on multi-view multi-label classification and on scenarios where both views and labels are missing (incomplete multi-view partial multi-label classification, IMVPMLC).

2.1 Multi-view multi-label classification

Multi-view multi-label classification aims to leverage the consistency and complementarity among different views

while modeling complex multi-label semantics, where each sample may be associated with multiple non-mutually-exclusive labels. Correlation modeling is a classical idea in multi-view learning, in which Canonical Correlation Analysis (CCA) and its variants learn a shared representation space by maximizing the correlation between projected representations from different views, and have been widely used in cross-view and cross-modal retrieval tasks (Yu et al. 2019; Shu and Zhao 2021). In addition to correlation modeling, matrix factorization and latent semantic modeling methods are also commonly used to explore the relationships between view features and label semantics. For example, LSA-MML (Zhang et al. 2018) exploits complementary information across different views through matrix factorization and latent semantic alignment, thereby improving classification performance; LVSL (Zhao et al. 2023a) further considers view misalignment and missing labels by learning view-specific labels and fusing them via low-rank label correlations. However, such methods typically rely on linear or low-rank assumptions, which limits their expressive power in capturing complex nonlinear cross-view relationships. In recent years, deep representation learning has also advanced related multi-view representation learning and multi-label prediction methods. For instance, MFLVC (Xu et al. 2022) combines multi-level feature learning with contrastive learning to jointly optimize feature reconstruction and cross-view consistency, while KSSNet (Wang et al. 2020) models statistical and knowledge structures through label graphs and promotes interactive learning between features and labels. Although these methods improve multi-view representation learning or multi-label prediction from different perspectives, most of them are still designed for scenarios where views and labels are relatively complete. When view-level and label-level incompleteness occur simultaneously, how to recover missing view information, suppress noise introduced during completion, and exploit potential supervisory signals from unknown labels remains a challenging problem.

2.2 Incomplete multi-view multi-label classification

In practical applications, multi-view multi-label classification often presents a more realistic yet challenging scenario where data is intrinsically incomplete, with both views and labels frequently absent owing to collection difficulties and annotation constraints. To address the challenges posed by this dual deficiency, researchers have proposed several representative methods. For view incompleteness, operating under the shared subspace assumption, the MVL-IV algorithm by Xu et al. (2015) restores missing views by strategically leveraging the inherent complementarity and consistency among the available views. For label incompleteness, Zhu et al. proposed a unified framework called GLOCAL, which simultaneously explores the correlation between global and local labels, jointly restores missing labels, and trains classifiers (Zhu et al. 2018). Moreover, deep learning has emerged as the dominant paradigm for feature extraction, largely due to its proven superiority in learning hierarchical and highly discriminative representations directly from data. The DD-IMvMLC-net framework (Wen et al. 2024) leverages an end-to-end deep architecture that seamlessly integrates view-specific encoders and decoders with weighted fusion and classification components. This integrated design enables robust modeling and accurate prediction under the dual incompleteness of views and labels across both semi-supervised and supervised settings. Liu et al. introduced the Transformer structure into incomplete multi-view multi-label tasks, using masked attention and weighted loss to handle missing views and labels respectively (Liu et al. 2023d).

Although existing methods have made progress in handling incomplete views or missing labels, many of them still rely on view and label indicator matrices to filter unavailable entries during learning. Such a strategy is useful for avoiding invalid supervision, but it does not fully recover the latent semantic information carried by missing views, nor does it explicitly distinguish the reliability of original and imputed information. In addition, the potential supervisory signals in unknown label positions are often insufficiently exploited, especially under label sparsity and class imbalance. To address these limitations, our framework proactively learns robust feature recovery through graph convolution, introduces a mask-gated dual-path strategy to separate high- and low-reliability information sources, and employs category-aware adaptive supervision to compensate for label scarcity.

3 The proposed method

This section presents the proposed method from three perspectives: Graph-guided Cross-view Completion, Dual-Path Multi-View Fusion, and Category-Adaptive Pseudo-Label Augmentation. The overall architecture of DGGI-Net is illustrated in Fig. 1. As shown in Fig. 1, DGGI-Net first recovers missing views through graph-guided cross-view completion, and then forms two fusion paths under the guidance of the view-availability mask. The high-quality path is used for final prediction, whereas the low-quality path provides candidate predictions for category-adaptive pseudo-label augmentation. In addition, the contrastive objective and neighborhood regularization are incorporated to improve the semantic consistency and structural reliability of the recovered views.

3.1 Problem definition

We formulate the considered task as follows. Suppose that the dataset contains n samples, and each sample is described

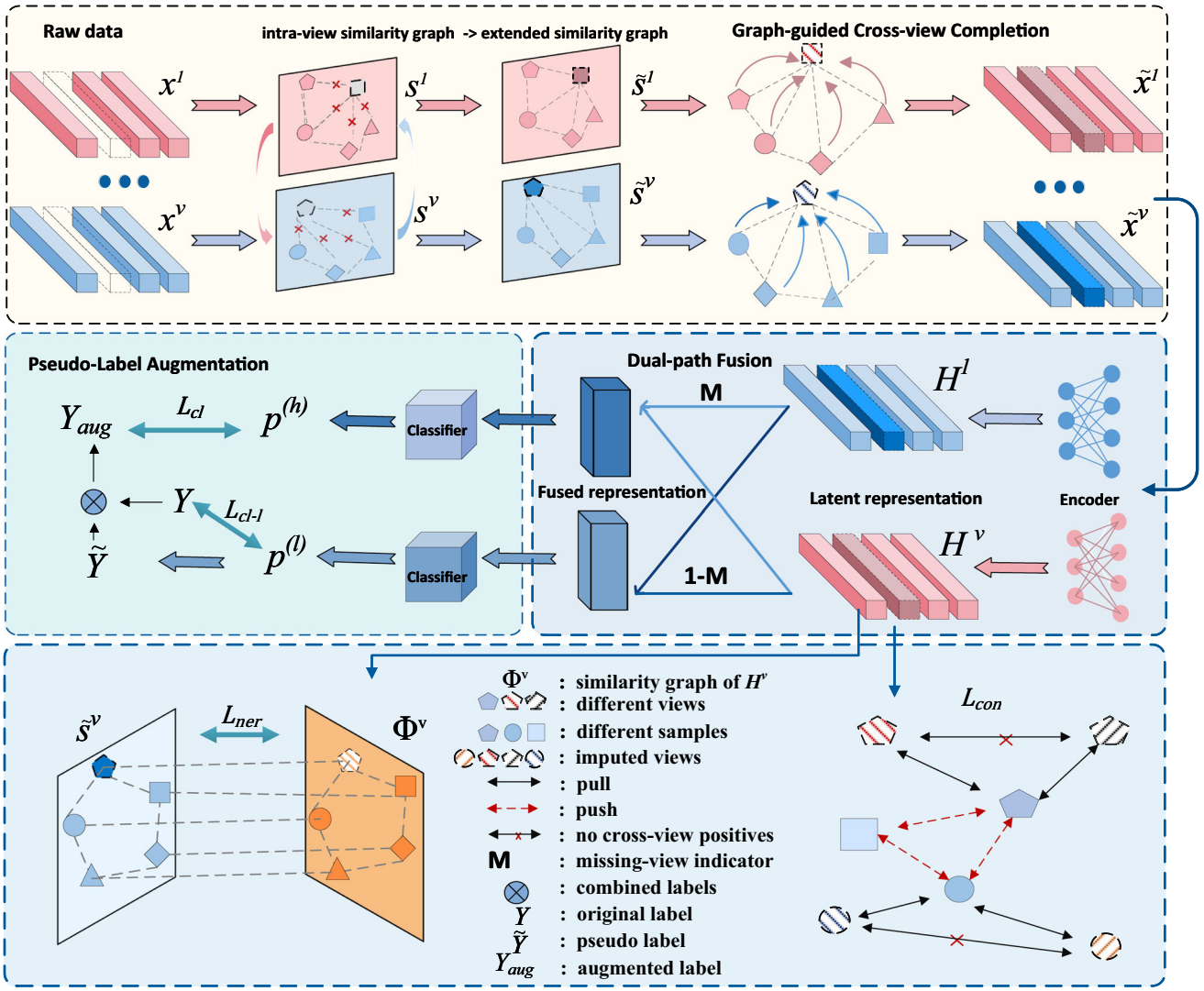


Fig. 1 Overall architecture of DGGI-Net. The framework first constructs extended similarity graphs and performs graph-guided cross-view completion to recover missing views. The completed features are then encoded and routed into a high-quality fusion path and a low-quality fusion path according to the view-availability mask. The high-quality path produces the final prediction, while the low-quality path provides candidate predictions for CAPLA. The contrastive objective and neighborhood regularization are further introduced to enhance the semantic consistency and structural reliability of the recovered views

by m views. For the v-th view, the corresponding feature matrix is denoted by $\mathbf{X}^{(v)} \in \mathbb{R}^{n \times d_v}$, while the ground-truth label matrix is denoted by $\mathbf{Y} \in \{0,1\}^{n \times c}$. To characterize dual incompleteness, we further introduce two binary mask matrices: a view-observation mask $\mathbf{M} \in \{0,1\}^{n \times m}$, where $M_{i,v} = 0$ indicates that the v-th view of the i-th sample is missing, and a label-observation mask $\mathbf{G} \in \{0,1\}^{n \times c}$, where $G_{i,j} = 0$ means that the j-th label of the i-th sample is unobserved. The goal is to learn a classifier $f : (\mathbf{X}^{(1)}, \ldots, \mathbf{X}^{(m)}) \mapsto \hat{\mathbf{Y}} \in [0,1]^{n \times c}$, which can accurately predict the complete label matrix $\mathbf{Y}$ under the joint incompleteness of views and labels. For convenience, the main notations used throughout the paper are summarized in Table 1.

3.2 Graph-guided cross-view completion

The inherent challenge in incomplete multi-view data arises from missing observation dimensions, which can induce models to learn biased joint probability distributions, thereby undermining their generalization capability in downstream tasks (Liu et al. 2024b). A natural way to address this issue is that proper completion of missing views can substantially improve model performance. Inspired by Dong et al. (2025), we propose a graph-guided cross-view completion (GCC) module for high-fidelity imputation of missing views. The module operates via two complementary components: (1) dynamically constructing multi-level adjacency matrices to capture complex intra- and cross-view relationships; (2)

Table 1 Main notations of this paper

Symbol	Description
n	Number of samples.
c	Number of classes.
m	Number of views.
d_v	Feature dimension of the v-th view.
$X^{(v)}$	Feature matrix of view v, $X^{(v)} \in \mathbb{R}^{n\times d_v}$.
Y	Ground-truth label matrix, $Y \in \{0,1\}^{n\times c}$.
M	Missing-view indicator, $M \in \{0,1\}^{n\times m}$.
G	Label-availability mask, $G \in \{0,1\}^{n\times c}$.
$S^{(v)}$	Intra-view similarity matrix, $S^{(v)} \in \mathbb{R}^{n\times n}$.
$\tilde{S}^{(v)}$	Extended similarity matrix of view v.
$\tilde{X}^{(v)}$	Completed feature matrix of view v.
$H^{(v)}$	Latent representation of view v, $H^{(v)} \in \mathbb{R}^{n\times d_z}$.
$\Phi^{(v)}$	Similarity matrix of $H^{(v)}$.
$z^{(h)}$, $z^{(l)}$	Fused features from high- and low-quality fusion paths.
$p^{(h)}$, $p^{(l)}$	Predictions from high- and low-quality fusion paths.
η_1, η_0	Relaxation coefficients for positive/negative thresholds.
τ	Temperature in contrastive loss.
γ_k, ρ_k	Global positive/negative ratios for class k.
τ_k^+, τ_k^-	Class-aware positive/negative thresholds for class k.
$\tilde{Y}$	Pseudo-label matrix, $\tilde{Y} \in \{-1,0,1\}^{n\times c}$.
Y_{aug}, G_{aug}	Augmented label matrix and availability mask.
$\mathcal{L}_{\text{cls}}$	Primary classification loss (high-quality path).

performing graph-convolutional feature propagation on the learned similarity graph to achieve high-precision completion of missing features. The multi-level adjacency matrix is employed to adaptively capture sample similarity for view recovery. For each view $v \in \{1,2,\ldots,m\}$, the intra-view similarity matrix is first computed as:

$$S_{ij}^{(v)} = \begin{cases} \dfrac{\langle x_i^{(v)}, x_j^{(v)} \rangle}{\|x_i^{(v)}\|_2 \, \|x_j^{(v)}\|_2}, & \text{if } M_{i,v}M_{j,v} = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

On this basis, we obtain the extended similarity matrix $\tilde{S}^{(v)}$ for each view by filling the missing entries with cross-view consensus:

$$\tilde{S}_{ij}^{(v)} = \begin{cases} S_{ij}^{(v)}, & \text{if } M_{i,v}M_{j,v} = 1, \\ \dfrac{\sum_{u=1}^{m} M_{i,u}M_{j,u}\, S_{ij}^{(u)}}{\sum_{u=1}^{m} M_{i,u}M_{j,u}}, & \text{if } M_{i,v}M_{j,v} = 0 \text{ and } \sum_{u=1}^{m} M_{i,u}M_{j,u} > 0, \\ 0, & \text{if } M_{i,v}M_{j,v} = 0 \text{ and } \sum_{u=1}^{m} M_{i,u}M_{j,u} = 0 \end{cases} \tag{2}$$

This design enables $\tilde{S}^{(v)}$ to fill missing regions based on cross-view consensus while preserving view-specific structures, so that even if a sample is completely absent in view v, its neighborhood relationships in the extended graph of view v can still be inferred from other available views. Leveraging the extended similarity graph $\tilde{S}^{(v)}$, we employ a graph-convolutional operation for feature reconstruction. A key advantage of graph convolution lies in its ability to aggregate neighborhood features according to sample-wise similarity (Wang et al. 2023, 2024b). Accordingly, for a target sample i whose v-th view is missing, its imputed feature is defined as

$$\hat{x}_i^{(v)} = \text{ReLU}\left(W \sum_{q\in\mathcal{N}_K^{(v)}(i)} \frac{\tilde{S}_{i,q}^{(v)}}{\sum_{r\in\mathcal{N}_K^{(v)}(i)} \tilde{S}_{i,r}^{(v)}} x_q^{(v)} + b \right) \tag{3}$$

where $\mathcal{N}_K^{(v)}(i)$ denotes the set of K nearest *available* neighbors of sample i in view v (i.e., $M_{q,v} = 1$) according to $\tilde{S}^{(v)}$, and W and b are the learnable weight matrix and bias vector, respectively.

Finally, for each view v, the completed feature matrix $\tilde{X}^{(v)}$ is formed by retaining the original features of available samples ($M_{i,v} = 1$) and replacing the features of missing samples ($M_{i,v} = 0$) with their imputed results $\hat{x}_i^{(v)}$:

$$\tilde{x}_i^{(v)} = \begin{cases} x_i^{(v)}, & \text{if } M_{i,v} = 1, \\ \hat{x}_i^{(v)}, & \text{if } M_{i,v} = 0, \end{cases} \qquad i = 1,\ldots,n. \tag{4}$$

In this module, we impute the missing views and obtain the completed feature matrices $\{\tilde{X}^{(v)}\}_{v=1}^m$ together with the extended similarity graphs $\{\tilde{S}^{(v)}\}_{v=1}^m$. In the subsequent modules, $\tilde{S}^{(v)}$ will serve as a structural anchor to suppress geometric distortions induced by graph-convolutional aggregation and to maintain spatial structural consistency, whereas the completed features $\tilde{X}^{(v)}$ will act as the core inputs for subsequent feature fusion and label prediction.

3.3 Dual-path multi-view fusion

Although the completion module in Section 3.2 recovers missing views with relatively high quality, the imputed features inevitably contain noise and uncertainty due to

approximation errors in neighbor selection and aggregation, as well as potential semantic drift (Lin et al. 2021). In contrast, features from originally available views derive directly from real observations and therefore exhibit higher fidelity. If the two sources are fused indiscriminately, noise from imputed views can contaminate high-fidelity signals and degrade overall predictive performance (Xie et al. 2023).

To fully leverage complementary information from imputed views while preventing feature cross-contamination, we propose a dual-path fusion strategy. Its core idea is to employ the view-availability mask M as a gating mechanism to regulate information flow, enabling the model to process features from different sources through logically separated paths within a shared encoding framework.

First, we employ a view-specific encoder $\Psi^{(v)}$ for each view v to project the completed features $\tilde{X}^{(v)}$ into a latent representation. Let $h_i^{(v)}$ denote the i-th row of $H^{(v)}$.

$$\left\{\Psi_{\theta^{(v)}}^{(v)} : \tilde{X}^{(v)} \in \mathbb{R}^{n\times d_v} \to H^{(v)} \in \mathbb{R}^{n\times d_z}\right\}_{v=1}^{m} \tag{5}$$

The high-quality fusion path $z^{(h)}$ aggregates only features from originally available views. We apply the view-availability mask M to gate the inputs so that the fused representation consists of high-fidelity signals, providing a reliable foundation for final classification:

$$z_i^{(h)} = \frac{\sum_{v=1}^{m} M_{i,v}\cdot h_i^{(v)}}{\sum_{v=1}^{m} M_{i,v}} \tag{6}$$

Conversely, the low-quality fusion path $z^{(l)}$ aggregates only features from imputed views (after completion) and is gated by the complement of the availability mask $1-M$. The fused representation from this path is used to generate high-confidence pseudo-labels, thereby indirectly improving overall classification performance:

$$z_i^{(l)} = \frac{\sum_{v=1}^{m}\left(1-M_{i,v}\right) h_i^{(v)}}{\sum_{v=1}^{m}\left(1-M_{i,v}\right) + \varepsilon}, \quad \varepsilon > 0. \tag{7}$$

where ε is a small smoothing constant to prevent division by zero.

To explicitly improve the quality of imputed views and enhance semantic consistency across views, we introduce a contrastive learning mechanism driven by view-consistency constraints. The contrastive loss $\mathcal{L}_{\text{con}}$ treats the representations of original views as reliable anchors and pulls the corresponding imputed views of the same sample toward them, encouraging the imputed features to learn the semantics of their original counterparts rather than reconstructed noise. Positive pairs follow the semantic equivalence principle: all originally available views of the same sample form positive pairs with each other, and each original view together with its imputed version is also regarded as a positive pair to guide the imputed features. We intentionally avoid pairing different imputed views of the same sample to prevent reinforcing reconstructed artifacts. Negative pairs are constructed from arbitrary view combinations belonging to different samples, strengthening cross-sample discriminability while refining the imputed representations.

For each view pair (p, q), we instantiate the contrastive objective using an InfoNCE form. The pair-wise contrastive loss is defined as:

$$\mathcal{L}_{\text{con}}^{(p,q)} = -\frac{1}{|\mathcal{P}_{p,q}|}\sum_{(i,p),(j,q)\in\mathcal{P}_{p,q}} \log\frac{\exp\left(s_{i,j}^{(p,q)}/\tau\right)}{\sum_{(k,u)\in\mathcal{A}_{p,q}(i)}\exp\left(s_{i,k}^{(p,u)}/\tau\right)} \tag{8}$$

where $\mathcal{P}_{p,q}$ denotes the set of positive pairs constructed between views p and q of the same sample, and $|\mathcal{P}_{p,q}|$ is its cardinality. The term $s_{i,j}^{(p,q)}$ denotes the similarity between latent view-level representations, i.e., $s_{i,j}^{(p,q)} = \text{sim}(h_i^{(p)}, h_j^{(q)})$, where $\text{sim}(\cdot,\cdot)$ is cosine similarity and τ is the temperature parameter. The index set $\mathcal{A}_{p,q}(i) = \{(j,q)\}\cup\mathcal{N}_{p,q}(i,p)$ contains the positive pair (j,q) together with the negatives $\mathcal{N}_{p,q}(i,p)$ associated with the anchor (i,p), i.e., view-sample pairs drawn from other samples or views.

The overall contrastive loss then aggregates contributions from all unordered view pairs:

$$\mathcal{L}_{\text{con}} = \sum_{1\le p<q\le m}\mathcal{L}_{\text{con}}^{(p,q)} \tag{9}$$

Furthermore, to preserve topological consistency during graph convolution operations, we introduce a neighborhood regularization loss $\mathcal{L}_{\text{ner}}$. Although graph convolution effectively aggregates neighborhood information for view imputation, successive convolutions may distort the original feature-space topology. Such distortion can compromise the structural consistency captured by the extended similarity matrices $\tilde{S}^{(v)}$ and impair downstream performance. The neighborhood regularization term enforces geometric alignment between the encoded features and the extended similarity graphs. Specifically, we construct a cosine-similarity matrix $\Phi^{(v)} \in \mathbb{R}^{n\times n}$ whose entries are given by $\Phi_{ij}^{(v)} = \text{sim}\left(h_i^{(v)}, h_j^{(v)}\right)$, where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, $h_i^{(v)}$ is the i-th row of $H^{(v)}$.

To preserve structural consistency, we enforce global cosine alignment between the vectorized forms of $\Phi^{(v)}$ and

$\tilde{S}^{(v)}$ by minimizing the following loss:

$$\mathcal{L}_{\text{ner}} = \sum_{v=1}^{m}\Big(1 - \frac{\text{vec}\big(\Phi^{(v)}\big)^{\top}\text{vec}\big(\tilde{S}^{(v)}\big)}{\big\|\text{vec}\big(\Phi^{(v)}\big)\big\|_2\,\big\|\text{vec}\big(\tilde{S}^{(v)}\big)\big\|_2}\Big) \tag{10}$$

where vec$(\cdot)$ stacks a matrix into a column vector. By minimizing $\mathcal{L}_{\text{ner}}$, we encourage the latent representations to retain the original inter-sample similarity structure, thereby suppressing geometric distortions from convolutional aggregation and enhancing consistency between imputed features and the underlying topology.

Finally, the fused representations from each path are fed into two independent prediction heads to generate predictions:

$$p_i^{(h)} = \sigma\big(R^{(h)}(z_i^{(h)})\big), \qquad p_i^{(l)} = \sigma\big(R^{(l)}(z_i^{(l)})\big) \tag{11}$$

where $\sigma(\cdot)$ denotes the sigmoid activation. The predictions from the high-quality fusion path $p_i^{(h)}$ serve as the final output, while those from the low-quality fusion path $p_i^{(l)}$ are used for pseudo-label generation in the category-adaptive pseudo-label augmentation module.

3.4 Category-adaptive pseudo-label augmentation

In multi-label learning, missing labels bias the observed label distribution, disrupt inter-label correlations, and degrade generalization. To address this, we propose Category-Adaptive Pseudo-label Augmentation (CAPLA). We first estimate global class statistics from the observed labels, and then filter high-confidence pseudo-labels from the low-quality path to supervise the high-quality path. The design follows two principles: (1) exploit the complementary information carried by imputed views while decoupling them from the high-quality path to avoid noise contamination; and (2) augment supervision with high-confidence pseudo-labels to alleviate label scarcity.

Due to reconstruction noise and inherently imbalanced label distributions, generating reliable pseudo-labels from the low-quality path is challenging. Fixed-threshold strategies are suboptimal: overly strict thresholds miss true positives of rare classes, whereas overly lenient thresholds yield excessive false positives in frequent classes. Accordingly, CAPLA adopts category-adaptive thresholds derived from global label statistics. Figure 2 illustrates the pseudo-label generation process of CAPLA. CAPLA uses the observed labels to estimate class-wise label statistics, and then applies category-adaptive thresholds to the low-quality path predictions to assign reliable pseudo-labels or abstention states, which are further used to augment the supervision of the high-quality path.

We estimate the global positive ratio γ_k and negative ratio ρ_k for each class $k \in \{1, \ldots, c\}$ from available labeled data:

$$\gamma_k = \frac{\sum_{i=1}^{n}\mathbb{I}(Y_{i,k} = 1 \wedge G_{i,k} = 1)}{\sum_{i=1}^{n}\mathbb{I}(G_{i,k} = 1)}, \quad \rho_k = 1 - \gamma_k \tag{12}$$

where $\mathbb{I}(\cdot)$ is the indicator function. γ_k estimates the conditional positive ratio for class k, i.e., the proportion of positive samples among those whose label k is observed.

Using these statistics, CAPLA generates category-specific confidence thresholds τ_k^+ and τ_k^- from $p^{(l)}$. To balance recall with precision and ensure stable training, we employ a progressive relaxation strategy:

$$\eta_1 = \alpha \cdot \frac{t}{T}, \quad \eta_0 = \beta \cdot \frac{t}{T}, \quad \alpha, \beta \in [0, 1]. \tag{13}$$

where t is the current epoch and T is the total number of training epochs. The factor t/T ensures that the thresholds adapt gradually from stricter to looser as training progresses, enhancing early-stage stability and recall in later stages. The hyperparameters α and β control the relaxation rates of positive and negative thresholds, respectively; their sensitivity is analyzed in Section 4.4.

The thresholds are defined as empirical quantiles:

$$\tau_k^+ = \text{quantile}\Big(\{p_{j,k}^{(l)}\}_{j=1}^{n},\ 1 - \eta_1\gamma_k\Big) \tag{14}$$

$$\tau_k^- = \text{quantile}\Big(\{p_{j,k}^{(l)}\}_{j=1}^{n},\ \eta_0\rho_k\Big) \tag{15}$$

where quantile$(\mathcal{S}, q)$ returns the value at the q-th quantile of the set $\mathcal{S}$ for $q \in [0, 1]$. Since the quantile levels depend on γ_k and ρ_k, the thresholds are class-aware and reflect the empirical label frequencies, allowing pseudo-label selection to adapt to the observed label distribution instead of being governed by a single global threshold.

Based on these thresholds, the pseudo-label $\tilde{y}_{j,k}$ for unlabeled sample j and class k is then assigned by

$$\tilde{y}_{j,k} = \begin{cases} 1, & p_{j,k}^{(l)} > \tau_k^+, \\ 0, & p_{j,k}^{(l)} < \tau_k^-, \\ -1, & \text{otherwise (abstain)} \end{cases} \tag{16}$$

The resulting matrix $\tilde{Y} = \big[\tilde{y}_1, \ldots, \tilde{y}_n\big]^{\top} \in \{-1, 0, 1\}^{n\times c}$ respects the estimated class distribution while suppressing unreliable supervision. A value of -1 denotes *abstention*: for low-confidence sample-class pairs we refrain from assigning a pseudo-label. This allocation rule has two key benefits: (i) for each class k, the fraction of positive pseudo-labels is encouraged to remain consistent with the estimated global ratio γ_k, reducing additional distribution shift introduced by pseudo-labeling; and (ii) by abstaining on marginal predictions ($\tau_k^- < p_{j,k}^{(l)} < \tau_k^+$), it filters low-confidence cases,

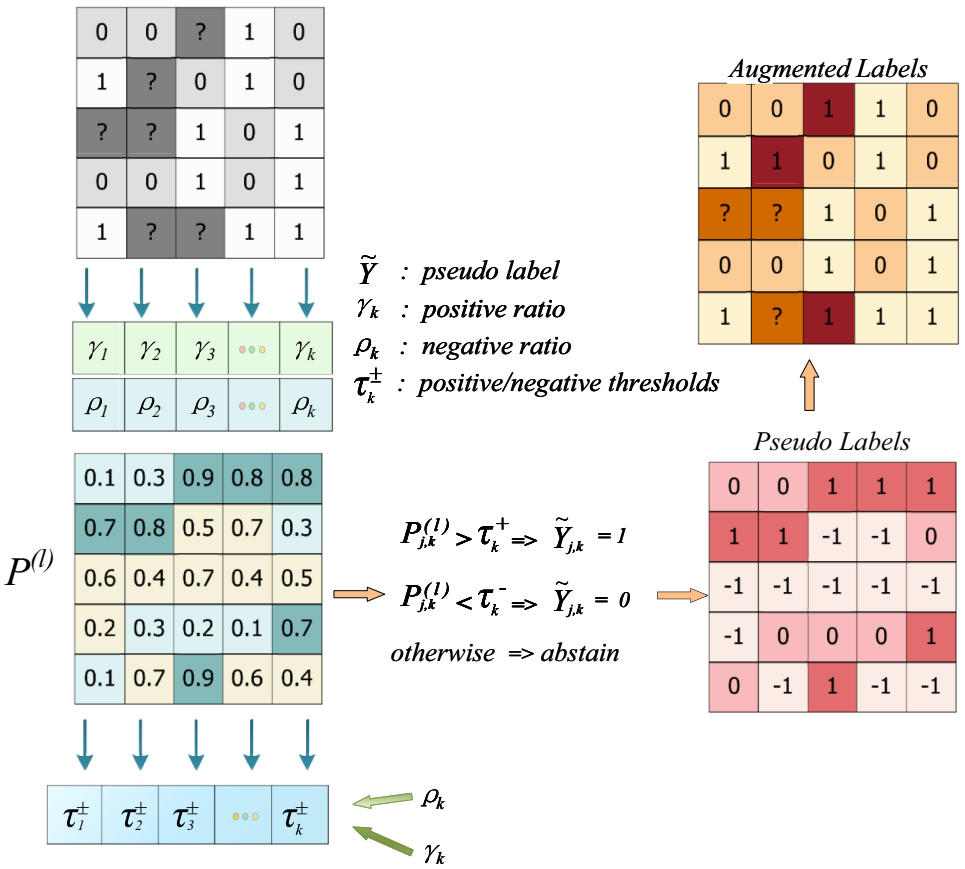


Fig. 2 The pipeline of category-adaptive pseudo-label augmentation in CAPLA. Based on the observed labels, CAPLA first estimates the class-wise positive and negative ratios γ_k and ρ_k. Guided by these statistics, predictions from the low-quality fusion path $p^{(l)}$ are processed with dynamically calibrated category-aware thresholds (τ_k^{+} and τ_k^{-}). Each unknown label entry is assigned a positive pseudo-label, a negative pseudo-label, or an abstention state according to its prediction confidence. The selected high-confidence pseudo-labels are then combined with the original observed labels to form augmented supervision

reduces reconstruction noise, and improves training stability without manual threshold tuning.

After generating the pseudo-label matrix $\tilde{Y}$ through CAPLA, we integrate it with ground-truth labels Y to form a consolidated supervision signal Y_{aug} for the high-quality path. Mathematically, the augmented label matrix Y_{aug} is constructed as:

$$Y_{\text{aug}} = Y \odot G + \mathbb{I}(\tilde{Y} \neq -1) \odot \tilde{Y} \odot (\mathbf{1} - G) \tag{17}$$

The indicator term $\mathbb{I}(\tilde{Y} \neq -1)$ explicitly filters out abstained pseudo-labels, so that only confident pseudo-labels at originally unobserved label positions are incorporated into the augmented supervision. Here, the operator $\odot$ denotes the element-wise (Hadamard) product, which ensures that the integration is performed independently for each element in the matrices.

Simultaneously, the label availability mask is extended to include confident pseudo-labels:

$$G_{\text{aug}} = G + \mathbb{I}(\tilde{Y} \neq -1) \odot (\mathbf{1} - G) \tag{18}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function that returns 1 when the condition is satisfied and 0 otherwise. Thus, entries with $\tilde{Y}_{ik} = -1$ are excluded from both Y_{aug} and G_{aug}, and therefore do not contribute to the subsequent BCE loss.

To leverage the integrated supervisory signals, we compute the classification loss for the predictions of the high-quality path. This loss function is built upon the standard Binary Cross-Entropy (BCE) criterion, extended by incorporating the augmented mask matrix G_{aug} as instance-wise weights. This weighting scheme ensures the model focuses its learning on both the originally labeled instances and those newly assigned with high-confidence pseudo-labels. The loss is formally defined as:

$$\mathcal{L}_{cls} = -\frac{1}{\sum_{i=1}^{n}\sum_{k=1}^{c} G_{aug,ik}} \sum_{i=1}^{n}\sum_{k=1}^{c} G_{aug,ik}\Big[Y_{aug,ik}\log(p_{ik}^{(h)}) + (1 - Y_{aug,ik})\log(1 - p_{ik}^{(h)})\Big] \tag{19}$$

By supervising this path with the enriched label set (ground truth + high-confidence pseudo-labels), we directly enhance the discriminative power of the model's primary output while mitigating the risk of noise propagation from the imputed views used in the low-quality fusion path.

To keep the low-quality fusion path reliable as a pseudo-label source, we impose an auxiliary BCE loss on its predictions $p^{(l)}$, using only observed ground-truth labels

selected by the mask G. This stabilizes its discriminative capacity and supports subsequent high-confidence pseudo-label filtering:

$$\mathcal{L}_{cls-l} = -\frac{1}{\sum_{i=1}^{n}\sum_{k=1}^{c} G_{ik}} \sum_{i=1}^{n}\sum_{k=1}^{c} G_{ik}\Big[Y_{ik}\log(p_{ik}^{(l)}) + (1-Y_{ik})\log\big(1-p_{ik}^{(l)}\big)\Big] \tag{20}$$

The final optimization objective combines the core classification losses with the auxiliary structural constraints introduced earlier, forming a synergistic multi-task learning framework:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \mathcal{L}_{cls-l} + \lambda_1\mathcal{L}_{con} + \lambda_2\mathcal{L}_{ner} \tag{21}$$

This unified loss function jointly optimizes the key objectives of the DGGI-Net framework: accurate label prediction using reliable features $\mathcal{L}_{cls}$, stabilizing the pseudo-label generator L_{cls-l}, enforcing cross-view semantic consistency $\lambda_1\mathcal{L}_{con}$, and preserving the topological structure during view completion $\lambda_2\mathcal{L}_{ner}$. The hyperparameters λ_1,λ_2 balance the contribution of each component towards the overall goal of robust multi-view multi-label classification under dual incompleteness.

Algorithm 1 DGGI-Net Training Process for Dual-Incomplete MVMLC

Require: Dual-incomplete data $\{X^{(v)}\}_{v=1}^{m}$, view mask M, labels Y, label mask G, loss weights λ_1, λ_2, epochs T.
Ensure: Trained model parameters, final predictions $p^{(h)}$, and completed views $\{\tilde{X}^{(v)}\}_{v=1}^{m}$.
Initialize view encoders $\{\Psi^{(v)}\}_{v=1}^{m}$ and prediction heads $R^{(h)}$, $R^{(l)}$.
for $t = 1$ to T **do**
  for $v = 1$ to m **do**
    Compute $S^{(v)}$ and $\tilde{S}^{(v)}$ via Eqs. (1)–(2).
    Impute missing features $\hat{X}^{(v)}$ via Eq. (3), and form $\tilde{X}^{(v)}$ via Eq. (4).
    $H^{(v)} \leftarrow \Psi^{(v)}(\tilde{X}^{(v)})$ ▷ Encode
  end for
  Compute neighborhood regularization $\mathcal{L}_{\text{ner}}$ via Eq. (10).
  Compute contrastive loss $\mathcal{L}_{\text{con}}$ via Eqs. (8)–(9).
  Obtain dual-path features $z^{(h)}$, $z^{(l)}$ and predictions $p^{(h)}$, $p^{(l)}$ via Eqs. (6), (7), and (11).
  Update the schedules (η_1, η_0) via Eq. (13), and derive the thresholds τ_k^+, τ_k^- via Eqs. (14)–(15).
  Assign the pseudo-label matrix $\tilde{Y}$ via Eq. (16), and construct the augmented label matrix Y_{aug} and availability mask G_{aug} via Eqs. (17)–(18).
  Compute primary loss $\mathcal{L}_{\text{cls}}$ on $p^{(h)}$ and auxiliary loss $\mathcal{L}_{\text{cls-}l}$ on $p^{(l)}$ via Eqs. (19)–(20).
  Update all parameters by minimizing $\mathcal{L}_{\text{total}}$ via Eq. (21).
end for

4 Experiments

In this section, we first describe the experimental setup, then present and analyze the results, followed by ablation studies to assess the contribution of each module, and finally conduct a sensitivity analysis of key hyperparameters.

4.1 Experimental settings

4.1.1 Datasets

We evaluate the proposed DGGI-Net on five widely used multi-view multi-label image datasets, namely Corel5k (Duygulu et al. 2002), Pascal07 (Everingham et al. 2010), ESPGame (Ahn and Dabbish 2004), IAPRTC12 (Escalante et al. 2010), and Mirflickr (Huiskes and Lew 2008). For each sample, six visual features (GIST, HSV, Hue, SIFT, RGB, and LAB) are extracted as multi-view inputs. The statistics and brief descriptions of these datasets are summarized in Table 2 and outlined below.

- **Corel5k**: Contains 4,999 natural scene images annotated with 260 semantic labels. The label distribution is sparse and each image is usually associated with multiple labels, making it a common benchmark for evaluating the generalization ability of multi-label learning methods.
- **Pascal07**: Built upon PASCAL VOC 2007, this dataset consists of 9,963 images and 20 object categories. Each image often contains multiple objects and complex backgrounds, and it is a typical benchmark for multi-object, multi-label recognition.
- **ESPGame**: Consists of 20,770 images and 268 labels collected from an online annotation game. The labels are abundant and semantically diverse, which is suitable for assessing the modeling capability of methods under large-scale and diverse label spaces.
- **IAPRTC12**: Contains 19,627 images annotated with 291 labels, covering a wide range of scenes and object concepts. It is often used to evaluate how well a model captures fine-grained semantics and inter-label correlations.
- **Mirflickr**: Comprises 25,000 web images from Flickr annotated with 38 frequently used tags. The image content is highly diverse, and the dataset is widely adopted to examine model performance on open-domain image retrieval and multi-label prediction tasks.

4.1.2 Incomplete dataset preparation

Following prior work (Tan et al. 2018; Li and Chen 2022; Liu et al. 2023b, 2025), we synthetically construct incomplete

Table 2 Statistics of the five multi-view multi-label image datasets used in our experiments

Dataset	#Samples	#Labels	#Views	Avg. labels/sample
Corel5k	4999	260	6	3.40
Pascal07	9963	20	6	1.47
ESPGame	20770	268	6	4.69
IAPRTC12	19627	291	6	5.72
Mirflickr	25000	38	6	4.72

multi-view and partial multi-label settings for a fair evaluation. Given a view-missing ratio φ, we randomly remove $\varphi\%$ of sample–view pairs in each view while ensuring that every sample retains at least one observed view. The feature vectors of removed views are replaced with zero vectors, and the resulting missing pattern is recorded in the indicator matrix M. Subsequently, given a label-missing ratio χ, we randomly mask $\chi\%$ of the label entries in Y to obtain partial labels; the unobserved entries are marked in the label-availability mask G, and only labels with $G_{ik} = 1$ contribute to supervision. Finally, each dataset is split into 70%/15%/15% training, validation, and test sets for model training, hyperparameter tuning, and evaluation.

4.1.3 Compared methods

To thoroughly evaluate the performance of our proposed method, we compare it with ten representative approaches across five benchmark datasets. Table 3 summarizes their capabilities in handling multi-view data, missing views, and incomplete labels. Specifically, the compared methods include the single-view multi-label baseline DM2L (Ma and Chen 2021), the complete multi-view multi-label method LVSL (Zhao et al. 2023a), and several representative methods for incomplete multi-view multi-label learning, including IMVWL (Tan et al. 2018), NAIM3L (Li and Chen 2022), MTD (Liu et al. 2023c), DICNet (Liu et al. 2023b), AIMNet (Liu et al. 2024a), MSLPP (Long et al. 2024), QARF-Net (Lu et al. 2025), and RANK (Liu et al. 2025).

Table 3 Comparison of the methods used. The 'M-V' column indicates whether the method is intended for multi-view data, while 'I-V' and 'I-L' refer to the capability of handling incomplete views and incomplete labels, respectively

Method	Source	M-V	I-V	I-L
DM2L	PR '21	×	×	×
LVSL	TMM '23	✓	×	×
IMVWL	IJCAI '18	✓	✓	✓
NAIM3L	TPAMI '22	✓	✓	✓
MTD	NeurIPS '23	✓	✓	✓
DICNet	AAAI '23	✓	✓	✓
AIMNet	AAAI '24	✓	✓	✓
MSLPP	NN '24	✓	✓	✓
QARF-Net	TCSVT '25	✓	✓	✓
RANK	TPAMI '25	✓	✓	✓

For the single-view method DM2L, we train one model per view and report the best-performing view selected on the validation set. For methods that assume complete labels, unknown labels are set to zero during training. For LVSL, which does not natively handle missing views or incomplete labels, we train on all available views after imputing missing views with mean features and setting unknown labels to zero. For the remaining multi-view methods that natively accommodate dual incompleteness, including IMVWL, NAIM3L, MTD, DICNet, AIMNet, MSLPP, QARF-Net, and RANK, we follow their original experimental protocols to ensure a fair comparison.

4.1.4 Evaluation metrics

Adhering to the standard evaluation protocol in multi-label classification (Tan et al. 2018; Li and Chen 2022; Liu et al. 2023b, 2025), we evaluate all methods using six widely adopted metrics: Average Precision (AP), Ranking Loss (RL), Hamming Loss (HL), Area Under the ROC Curve (AUC), One-error (OE), and Coverage (Cov). Among these, AP and AUC are benefit-type metrics, where higher values indicate better performance, whereas RL, HL, OE, and Cov are cost-type metrics, for which lower values are preferable. To facilitate straightforward comparison, we report the latter four in their complementary form (i.e., 1-RL, 1-HL, 1-OE, and 1-Cov), so that a higher value consistently reflects better performance for all metrics.

4.1.5 Implementation details

The implementation of DGGI-Net was carried out in Python using the PyTorch framework. For all datasets, the initial learning rate was fixed at 1 to maintain consistency in parameter tuning across different experiments. The training setup remained the same across all trials, with a batch size of 128, momentum set to 0.9, and optimization performed using stochastic gradient descent (SGD). In the view completion module, the k-nearest neighbor (k-NN) graph

was constructed with $k = 7$ in all experiments. All experiments were conducted on a Windows-based system equipped with an NVIDIA GeForce RTX 5060 Laptop GPU (8 GB VRAM) and an AMD Ryzen 9 8940HX processor with Radeon Graphics. To enhance statistical reliability and mitigate potential bias caused by random data partitioning, each experiment was repeated 10 times under different random splits, and the average results together with their standard deviations were reported.

4.2 Experimental results and analysis

In the evaluation under dual-incomplete scenarios, we comprehensively compared the performance of DGGI-Net and existing methods on five standard multi-view multi-label datasets. Table 4 reports the results of ten compared methods and our approach under the setting of 50% missing views, 50% missing labels, and 70% training samples. The standard deviations over multiple runs are shown as subscripts in each cell.

To further evaluate the robustness of DGGI-Net under varying degrees of incompleteness, we vary one missing rate while keeping the other fixed at 50%. Specifically, either the view-missing rate or the label-missing rate is gradually increased from 0% to 70%. We also include RANK as a representative competing method that natively handles both missing views and missing labels. The results on Corel5k and Pascal07 are shown in Figs. 3 and 4, respectively, where AP and AUC are reported as representative metrics to reflect label-ranking quality and overall discriminative ability.

We also carried out a comparison experiment on the complete dataset without missing views or labels. The performance comparison is shown in Fig. 5 through radar charts. The detailed analysis is as follows:

- As presented in Table 4, under the stringent conditions of 50% missing views, 50% missing labels, and only 70% of the training data available, DGGI-Net consistently outperforms all other methods across multiple benchmark datasets. Specifically, DGGI-Net achieves the highest scores across all six evaluation metrics. For example, on Corel5k, the AP reaches 0.435, surpassing the strongest baseline, RANK, by 1.87%. On Pascal07, DGGI-Net attains an AP of 0.573, outperforming the strongest baseline, QARF-Net, by 3.06%. These performance improvements are also observed in other metrics such as 1-RL and AUC, further demonstrating DGGI-Net's robustness in handling missing views and labels. The performance gains can be attributed to the effective integration of three key components: dynamic graph-guided view completion, dual-path multi-view fusion, and category-adaptive pseudo-label augmentation. These elements collectively enhance feature recovery, suppress noise propagation, and alleviate label sparsity, leading to superior overall performance.
- As shown in Figs. 3 and 4, on both Corel5k and Pascal07, model performance declines more significantly when the label-missing rate is fixed at 50% and the view-missing rate gradually increases. In contrast, when the view-missing rate is fixed at 50% and the label-missing rate gradually increases, the performance degradation is relatively smaller. This indicates that view missingness poses a greater challenge than label missingness, because missing views directly reduce the raw visual information available to the model and thus affect the quality of representation learning. Meanwhile, DGGI-Net consistently outperforms RANK under both settings and achieves better overall performance in terms of AP and AUC. These results further demonstrate that the proposed graph-guided view completion and category-adaptive pseudo-label augmentation mechanisms enhance the robustness of DGGI-Net under severe dual incompleteness.
- We further evaluated DGGI-Net on the complete dataset without missing views or labels to examine the effectiveness of its core learning framework under ideal conditions. In this setting, since all views are fully observed, the GCC branch no longer performs actual missing-view recovery and naturally degenerates into structure-preserving cross-view feature propagation over complete observations. Accordingly, the contrastive objective in Eq. (9) is instantiated using only fully observed cross-view representations, while the pseudo-label augmentation module is disabled because no missing labels need to be supplemented. Meanwhile, the core objective terms in Eqs. (10), (9), and (19) are retained to preserve structural consistency, enhance cross-view semantic alignment, and maintain classification learning under fully observed conditions. As shown in the radar charts in Fig. 5, the resulting model still outperforms all competing methods on all six evaluation metrics. These results indicate that, beyond its effectiveness in dual-incomplete scenarios, the core optimization design of DGGI-Net remains capable of learning robust and discriminative representations even when the data are fully observed.

4.3 Ablation experiments

To systematically validate the effectiveness of the key design choices in DGGI-Net, we conduct ablation experiments on Corel5k and Pascal07 under 50% missing views and 50% missing labels. The results are reported in Table 5. The ablation study consists of three parts. Table 5(A) presents a progressive module-and-loss ablation to analyze the contribution of each component. Table 5(B) further investigates the

Table 4 Experimental results of ten baseline methods and our approach on five datasets under 50% missing views and 50% missing labels; standard deviations are shown as subscripts. The best results are highlighted in bold, and the second-best results are underlined. RI denotes the relative improvement of DGGI-Net over the strongest baseline method

DATA	METRIC	DM2L	LVSL	IMVWL	NAIM3L	DICNet	AIMNet	MSLPP	MTD	QARF-Net	RANK	OURS	RI(%)
Corel5k	AP↑	$0.262_{0.005}$	$0.342_{0.004}$	$0.283_{0.008}$	$0.309_{0.004}$	$0.381_{0.004}$	$0.400_{0.010}$	$0.413_{0.008}$	$0.415_{0.008}$	$0.411_{0.008}$	$\underline{0.427}_{0.006}$	$\mathbf{0.435}_{0.011}$	1.87
	1-HL↑	$\underline{0.987}_{0.000}$	$\underline{0.987}_{0.000}$	$0.978_{0.000}$	$\underline{0.987}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	$\mathbf{0.988}_{0.000}$	0.00
	1-RL↑	$0.843_{0.002}$	$0.881_{0.003}$	$0.865_{0.005}$	$0.878_{0.002}$	$0.882_{0.004}$	$0.902_{0.002}$	$0.901_{0.003}$	$0.893_{0.004}$	$0.900_{0.004}$	$\underline{0.913}_{0.002}$	$\mathbf{0.915}_{0.003}$	0.22
	AUC↑	$0.845_{0.002}$	$0.884_{0.003}$	$0.868_{0.005}$	$0.881_{0.002}$	$0.884_{0.004}$	$0.905_{0.003}$	$0.903_{0.004}$	$0.896_{0.004}$	$0.902_{0.004}$	$\underline{0.916}_{0.003}$	$\mathbf{0.918}_{0.003}$	0.22
	1-OE↑	$0.295_{0.014}$	$0.391_{0.009}$	$0.311_{0.015}$	$0.350_{0.009}$	$0.468_{0.007}$	$0.475_{0.018}$	$0.490_{0.020}$	$0.491_{0.012}$	$0.492_{0.013}$	$\underline{0.499}_{0.012}$	$\mathbf{0.512}_{0.019}$	2.61
	1-Cov↑	$0.647_{0.005}$	$0.718_{0.006}$	$0.702_{0.008}$	$0.725_{0.005}$	$0.727_{0.011}$	$0.771_{0.005}$	$0.767_{0.080}$	$0.749_{0.009}$	$0.761_{0.009}$	$\underline{0.800}_{0.005}$	$\mathbf{0.802}_{0.006}$	0.25
Pascal07	AP↑	$0.471_{0.008}$	$0.504_{0.005}$	$0.437_{0.018}$	$0.488_{0.003}$	$0.505_{0.012}$	$0.548_{0.008}$	$0.544_{0.010}$	$0.551_{0.004}$	$\underline{0.556}_{0.007}$	$0.555_{0.010}$	$\mathbf{0.573}_{0.010}$	3.06
	1-HL↑	$0.928_{0.001}$	$0.930_{0.000}$	$0.882_{0.004}$	$0.928_{0.001}$	$0.929_{0.001}$	$0.931_{0.001}$	$0.932_{0.001}$	$0.932_{0.001}$	$0.931_{0.001}$	$\underline{0.933}_{0.001}$	$\mathbf{0.934}_{0.001}$	0.11
	1-RL↑	$0.761_{0.005}$	$0.806_{0.003}$	$0.736_{0.015}$	$0.783_{0.001}$	$0.783_{0.008}$	$\underline{0.831}_{0.004}$	$0.819_{0.006}$	$\underline{0.831}_{0.003}$	$0.828_{0.004}$	$0.830_{0.005}$	$\mathbf{0.847}_{0.005}$	1.93
	AUC↑	$0.779_{0.004}$	$0.832_{0.002}$	$0.767_{0.015}$	$0.811_{0.001}$	$0.809_{0.006}$	$\underline{0.851}_{0.004}$	$0.841_{0.004}$	$\underline{0.851}_{0.003}$	$0.847_{0.004}$	$\underline{0.851}_{0.006}$	$\mathbf{0.866}_{0.004}$	1.76
	1-OE↑	$0.420_{0.011}$	$0.419_{0.008}$	$0.362_{0.023}$	$0.421_{0.006}$	$0.427_{0.015}$	$0.461_{0.013}$	$0.462_{0.013}$	$0.459_{0.007}$	$\underline{0.471}_{0.011}$	$0.463_{0.016}$	$\mathbf{0.481}_{0.017}$	2.12
	1-Cov↑	$0.692_{0.004}$	$0.751_{0.003}$	$0.677_{0.015}$	$0.727_{0.002}$	$0.731_{0.006}$	$0.783_{0.004}$	$0.770_{0.008}$	$\underline{0.784}_{0.003}$	$0.780_{0.005}$	$0.783_{0.006}$	$\mathbf{0.802}_{0.005}$	2.30
ESPGame	AP↑	$0.212_{0.002}$	$0.285_{0.003}$	$0.244_{0.005}$	$0.246_{0.002}$	$0.297_{0.002}$	$0.305_{0.004}$	$0.310_{0.004}$	$0.306_{0.003}$	$0.307_{0.004}$	$\underline{0.314}_{0.004}$	$\mathbf{0.328}_{0.004}$	4.46
	1-HL↑	$\underline{0.982}_{0.000}$	$\mathbf{0.983}_{0.000}$	$0.972_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	$\mathbf{0.983}_{0.000}$	0.00
	1-RL↑	$0.781_{0.001}$	$0.829_{0.001}$	$0.808_{0.002}$	$0.818_{0.002}$	$0.832_{0.001}$	$0.846_{0.002}$	$0.843_{0.002}$	$0.837_{0.002}$	$0.842_{0.001}$	$\underline{0.848}_{0.002}$	$\mathbf{0.860}_{0.002}$	1.42
	AUC↑	$0.785_{0.001}$	$0.833_{0.002}$	$0.813_{0.002}$	$0.824_{0.002}$	$0.836_{0.001}$	$0.850_{0.002}$	$0.847_{0.002}$	$0.842_{0.002}$	$0.846_{0.001}$	$\underline{0.853}_{0.002}$	$\mathbf{0.863}_{0.002}$	1.17
	1-OE↑	$0.294_{0.006}$	$0.389_{0.004}$	$0.343_{0.013}$	$0.339_{0.003}$	$0.439_{0.007}$	$0.442_{0.011}$	$0.455_{0.010}$	$0.447_{0.009}$	$0.448_{0.009}$	$\underline{0.462}_{0.015}$	$\mathbf{0.472}_{0.010}$	2.16
	1-Cov↑	$0.488_{0.003}$	$0.567_{0.005}$	$0.548_{0.004}$	$0.571_{0.003}$	$0.593_{0.003}$	$0.624_{0.005}$	$0.619_{0.004}$	$0.602_{0.004}$	$\underline{0.651}_{0.004}$	$0.630_{0.005}$	$\mathbf{0.655}_{0.005}$	0.61
IAPRTC12	AP↑	$0.234_{0.003}$	$0.304_{0.004}$	$0.237_{0.003}$	$0.261_{0.001}$	$0.323_{0.001}$	$0.329_{0.005}$	$0.340_{0.005}$	$0.332_{0.003}$	$0.338_{0.004}$	$\underline{0.346}_{0.004}$	$\mathbf{0.356}_{0.004}$	2.89
	1-HL↑	$\underline{0.980}_{0.000}$	$\mathbf{0.981}_{0.000}$	$0.969_{0.000}$	$\underline{0.980}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	$\mathbf{0.981}_{0.000}$	0.00
	1-RL↑	$0.823_{0.002}$	$0.861_{0.002}$	$0.833_{0.002}$	$0.848_{0.001}$	$0.873_{0.001}$	$0.883_{0.003}$	$0.882_{0.002}$	$0.875_{0.002}$	$0.882_{0.002}$	$\underline{0.888}_{0.002}$	$\mathbf{0.899}_{0.002}$	1.24
	AUC↑	$0.825_{0.001}$	$0.863_{0.001}$	$0.835_{0.001}$	$0.850_{0.001}$	$0.874_{0.000}$	$0.885_{0.003}$	$0.883_{0.002}$	$0.876_{0.001}$	$0.883_{0.002}$	$\underline{0.889}_{0.002}$	$\mathbf{0.899}_{0.002}$	1.12
	1-OE↑	$0.340_{0.006}$	$0.429_{0.009}$	$0.352_{0.008}$	$0.390_{0.005}$	$0.468_{0.002}$	$0.459_{0.008}$	$0.475_{0.012}$	$0.467_{0.005}$	$0.471_{0.011}$	$\underline{0.485}_{0.006}$	$\mathbf{0.491}_{0.006}$	1.24
	1-Cov↑	$0.529_{0.004}$	$0.597_{0.004}$	$0.564_{0.005}$	$0.592_{0.004}$	$0.649_{0.001}$	$0.673_{0.006}$	$0.670_{0.006}$	$0.649_{0.004}$	$0.668_{0.005}$	$\underline{0.685}_{0.005}$	$\mathbf{0.715}_{0.005}$	4.38
Mirflickr	AP↑	$0.514_{0.006}$	$0.553_{0.002}$	$0.490_{0.012}$	$0.551_{0.002}$	$0.589_{0.005}$	$0.602_{0.004}$	$0.615_{0.004}$	$0.607_{0.004}$	$\underline{0.616}_{0.004}$	$0.602_{0.007}$	$\mathbf{0.622}_{0.004}$	0.97
	1-HL↑	$0.878_{0.001}$	$0.885_{0.001}$	$0.839_{0.002}$	$0.882_{0.001}$	$0.888_{0.002}$	$0.890_{0.001}$	$\underline{0.892}_{0.001}$	$0.891_{0.001}$	$\underline{0.892}_{0.001}$	$0.890_{0.001}$	$\mathbf{0.893}_{0.001}$	0.11
	1-RL↑	$0.831_{0.003}$	$0.856_{0.001}$	$0.803_{0.008}$	$0.844_{0.001}$	$0.863_{0.004}$	$0.873_{0.002}$	$0.879_{0.002}$	$0.875_{0.002}$	$\underline{0.880}_{0.003}$	$0.875_{0.003}$	$\mathbf{0.884}_{0.002}$	0.45
	AUC↑	$0.828_{0.003}$	$0.844_{0.001}$	$0.787_{0.012}$	$0.837_{0.001}$	$0.849_{0.004}$	$0.861_{0.001}$	$0.865_{0.002}$	$0.862_{0.002}$	$\underline{0.866}_{0.002}$	$0.860_{0.003}$	$\mathbf{0.870}_{0.001}$	0.46
	1-OE↑	$0.510_{0.008}$	$0.607_{0.004}$	$0.511_{0.022}$	$0.585_{0.003}$	$0.637_{0.007}$	$0.651_{0.006}$	$\underline{0.671}_{0.007}$	$0.655_{0.004}$	$0.662_{0.008}$	$0.653_{0.009}$	$\mathbf{0.672}_{0.010}$	0.15
	1-Cov↑	$0.604_{0.005}$	$0.636_{0.001}$	$0.572_{0.013}$	$0.631_{0.002}$	$0.652_{0.007}$	$0.671_{0.004}$	$0.679_{0.003}$	$0.676_{0.004}$	$\underline{0.683}_{0.003}$	$0.670_{0.004}$	$\mathbf{0.690}_{0.003}$	1.02

Note: RI is calculated as (OURS − Best baseline)/Best baseline × 100%. For rows where DGGI-Net ties with the strongest baseline, RI is reported as 0.00%

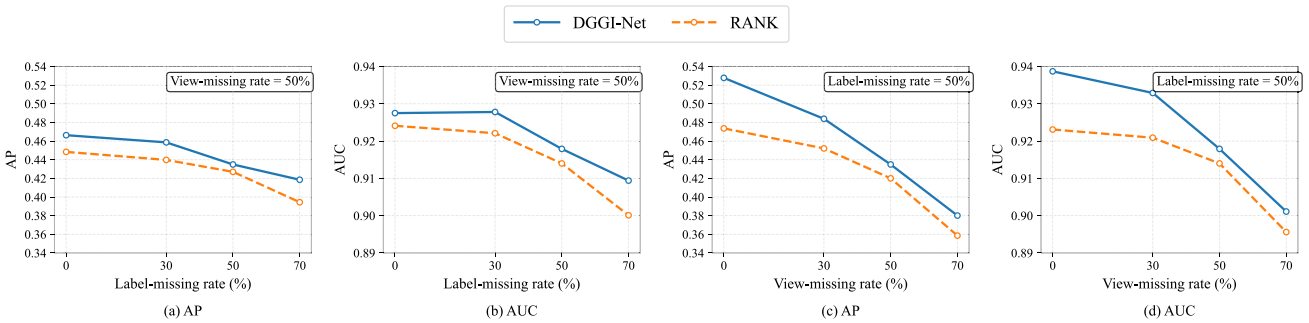


Fig. 3 Robustness comparison between DGGI-Net and RANK on the Corel5k dataset under varying incompleteness conditions. Panels (**a**)–(**b**) report AP and AUC when the label-missing rate varies while the view-missing rate is fixed at 50%. Panels (**c**)–(**d**) report AP and AUC when the view-missing rate varies while the label-missing rate is fixed at 50%

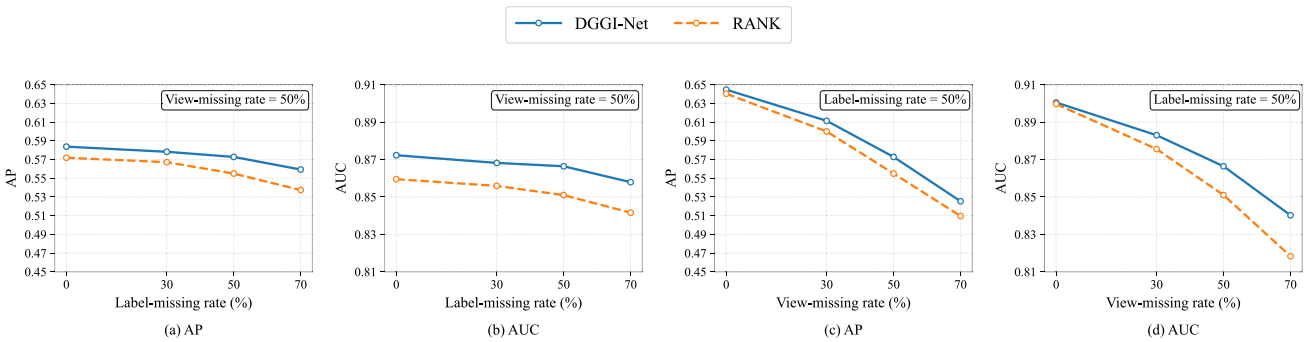


Fig. 4 Robustness comparison between DGGI-Net and RANK on the Pascal07 dataset under varying incompleteness conditions. Panels (**a**)–(**b**) report AP and AUC when the label-missing rate varies while the view-missing rate is fixed at 50%. Panels (**c**)–(**d**) report AP and AUC when the view-missing rate varies while the label-missing rate is fixed at 50%

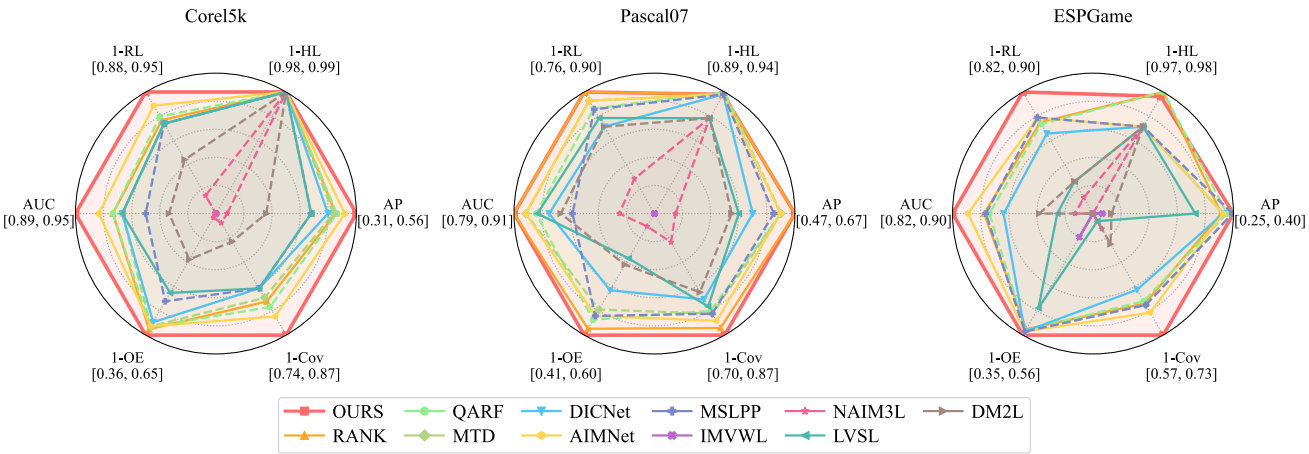


Fig. 5 Performance comparison on complete datasets using radar charts. The center indicates the lowest score, while the outer edges represent the best performance across six evaluation metrics

Table 5 Ablation study of DGGI-Net on Corel5k and Pascal07 under 50% missing views and 50% missing labels, including (A) module-and-loss ablation, (B) pseudo-label source ablation, and (C) max-pooling fusion ablation

Model Setting					Corel5k Dataset				Pascal07 Dataset			
Setting	$\mathcal{L}_{\text{cls}}$	$\mathcal{L}_{\text{ner}}$	$\mathcal{L}_{\text{con}}$	$\mathcal{L}_{\text{cls}-1}$	AP↑	1-HL↑	1-RL↑	AUC↑	AP↑	1-HL↑	1-RL↑	AUC↑
(A) Module-and-loss ablation												
Backbone	✓				0.399	0.987	0.894	0.897	0.546	0.931	0.823	0.846
$+\mathcal{L}_{\text{ner}}$	✓	✓			0.405	0.988	0.903	0.905	0.552	0.933	0.833	0.854
$+\mathcal{L}_{\text{con}}$	✓	✓	✓		0.426	0.988	0.915	0.917	0.570	0.934	0.843	0.864
$+\mathcal{L}_{\text{cls}-1}$ (Full)	✓	✓	✓	✓	**0.435**	**0.988**	**0.915**	**0.918**	**0.573**	**0.934**	**0.847**	**0.866**
(B) Pseudo-label source ablation												
CAPLA with $p^{(h)}$	✓	✓	✓	✓	0.432	0.988	0.915	0.917	0.572	0.933	0.845	0.864
(C) Max-pooling fusion ablation												
Max-pooling fusion	✓	✓	✓	✓	0.421	0.988	0.912	0.915	0.568	0.933	0.841	0.862

influence of the pseudo-label source in CAPLA. Table 5(C) compares different fusion aggregation strategies to examine the effectiveness of the mean-pooling design in Eqs. 6 and 7.

In Table 5(A), we use the Backbone model, which only adopts the primary classification loss $\mathcal{L}_{\text{cls}}$, as the baseline. To make the ablation settings directly aligned with the overall optimization objective, the table explicitly lists the loss terms enabled at each stage, including $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{ner}}$, $\mathcal{L}_{\text{con}}$, and $\mathcal{L}_{\text{cls}-1}$. Starting from the Backbone, we progressively introduce the graph-guided completion with neighborhood regularization, the cross-view contrastive loss, and CAPLA with the auxiliary low-quality path supervision. The results indicate that the incremental contributions of different components are not identical. Among them, $\mathcal{L}_{\text{con}}$ brings the most notable improvement, indicating that cross-view semantic alignment is particularly important for learning discriminative representations. Although the gains brought by $\mathcal{L}_{\text{ner}}$ and $\mathcal{L}_{\text{cls}-1}$ are relatively smaller in terms of AP, they still provide stable complementary benefits. Specifically, $\mathcal{L}_{\text{ner}}$ helps preserve neighborhood structures during view completion, while $\mathcal{L}_{\text{cls}-1}$ works with CAPLA to provide additional supervision for unknown label positions. Overall, these results suggest that $\mathcal{L}_{\text{con}}$ plays the dominant role in performance improvement, whereas $\mathcal{L}_{\text{ner}}$ and $\mathcal{L}_{\text{cls}-1}$ further enhance structural reliability and label-sparsity alleviation within the full framework.

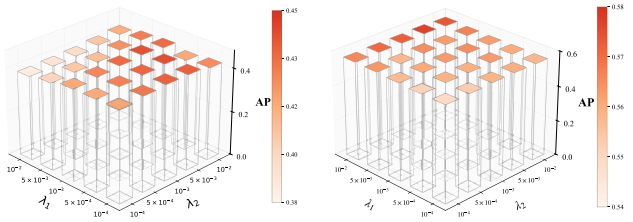
Table 5(B) further compares different pseudo-label sources. Specifically, we compare the full model, which uses the low-quality path prediction $p^{(l)}$ as the pseudo-label source, with a variant that replaces it with the high-quality path prediction $p^{(h)}$. The results show that using $p^{(h)}$ as the pseudo-label source leads to lower or no better performance than using $p^{(l)}$. This suggests that directly using the final prediction path $p^{(h)}$ to generate pseudo-labels may introduce a closed-loop self-training effect, where early prediction biases could be reinforced. In contrast, $p^{(l)}$ is generated from the imputation-dominant path and can provide complementary information different from that of the observed-view-dominant high-quality path. Meanwhile, the category-adaptive thresholds and abstention mechanism in CAPLA filter out low-confidence predictions. Therefore, $p^{(l)}$ is more suitable as an auxiliary pseudo-label source, while the final prediction is still produced by the more reliable high-quality path $p^{(h)}$.

Table 5(C) further compares the influence of different fusion aggregation strategies. Specifically, we replace the mean pooling in Eqs. 6 and 7 with max pooling. For the high-quality fusion path, max pooling can be formulated as $z_{i,d}^{(h)} = \max_{v:M_{i,v}=1} h_{i,d}^{(v)}$, which takes the maximum response of each latent dimension across the originally observed views of sample i. Similarly, for the low-quality fusion path, max pooling is performed across the imputed views on a dimension-wise basis. The results show that max pooling leads to lower overall performance than mean pooling. This indicates that although max pooling can emphasize strong local responses, it may also amplify noise or imputation errors from individual views. In contrast, mean pooling integrates information from multiple views and produces a smoother and more stable fused representation, making it more suitable for the proposed dual-path fusion framework.

4.4 Hyperparameter sensitivity analysis

We perform a sensitivity analysis to assess the influence of key hyperparameters in the DGGI-Net framework, including the loss weights λ_1 and λ_2, the pseudo-label relaxation coefficients α and β, and the contrastive temperature τ. We report Average Precision (AP) as the primary evaluation metric; other measures exhibit similar trends and are omitted for brevity.

First, we examine the influence of the loss weights λ_1 and λ_2 by varying their values within the range $[10^{-4}, 10^{-2}]$. As shown in Fig. 6, the model exhibits stable performance



(a) Corel5k (b) Pascal07

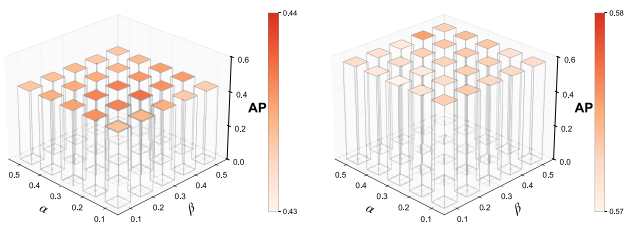
Fig. 6 The influence of penalty coefficients λ_1 and λ_2 on the Corel5k and Pascal07 datasets

across this interval, and the best results are obtained when both weights lie in a moderate range that balances the classification objective with the structural regularizers. Specifically, on Corel5k the best AP is achieved at $\lambda_1 = 10^{-3}$ and $\lambda_2 = 5 \times 10^{-3}$, while on Pascal07 the optimum appears near $\lambda_1 = 1 \times 10^{-2}$ and $\lambda_2 = 5 \times 10^{-3}$. This suggests that DGGI-Net does not rely on finely tuned values of λ_1 and λ_2 as long as they are set within a reasonable magnitude.

Next, we evaluate the pseudo-label relaxation coefficients α and β, sweeping both within [0.1, 0.5] (Fig. 7). On Corel5k, the highest AP is obtained at $\alpha = 0.2$ and $\beta = 0.3$, whereas Pascal07 peaks at $\alpha = 0.5$ and $\beta = 0.4$. Although the exact optimal values vary slightly between datasets, the performance curves remain relatively flat around these settings, indicating that CAPLA is robust to the precise choice of α and β as long as they fall in a moderate range.

Finally, we assess the contrastive temperature τ over the set $\{0.01, 0.05, 0.1, 0.2, 0.3\}$ (Fig. 8). The AP curve for Corel5k peaks at $\tau = 0.1$, while Pascal07 achieves its optimum at $\tau = 0.05$. Extremely small values of τ over-sharpen the similarity distribution and may destabilize optimization, whereas excessively large values tend to over-smooth similarities and weaken discriminative power. In practice, an intermediate value around the empirical optimum for each dataset yields consistently strong performance.

In summary, DGGI-Net demonstrates robust behavior across a wide range of hyperparameter configurations. We recommend setting λ_1, λ_2 in the range 10^{-3}-5×10^{-3}, choosing α, β from [0.1, 0.5], and using an intermediate temperature τ near its empirical optimum. Within these ranges, the model performance remains stable without requiring meticulous tuning.



(a) Corel5k (b) Pascal07

Fig. 7 Sensitivity to class-aware pseudo-label relaxation coefficients α and β

4.5 Pseudo-label behavior analysis

To further characterize the behavior of CAPLA and analyze its reliability, we conduct a pseudo-label analysis during training. Specifically, we report three metrics: Total Fill Rate (FR), Abstain Rate (AR), and Pseudo-label Accuracy (PLA). FR measures the fraction of originally unknown label entries that are assigned confident pseudo-labels, AR measures the fraction of unknown label entries that remain abstained, and PLA measures the consistency between the generated pseudo-labels and the hidden ground-truth labels. These metrics are defined as follows:

$$\mathrm{FR} = \frac{\sum_{i=1}^{n}\sum_{k=1}^{c}\mathbb{I}(G_{ik}=0)\mathbb{I}(\widetilde{Y}_{ik}\neq -1)}{\sum_{i=1}^{n}\sum_{k=1}^{c}\mathbb{I}(G_{ik}=0)},$$

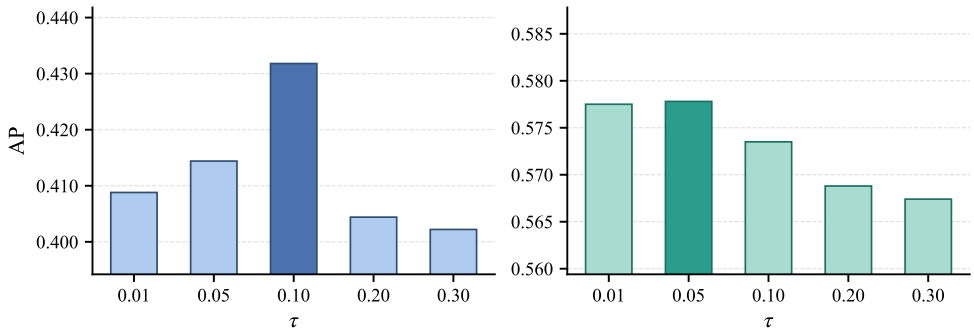
$$\mathrm{AR} = \frac{\sum_{i=1}^{n}\sum_{k=1}^{c}\mathbb{I}(G_{ik}=0)\mathbb{I}(\widetilde{Y}_{ik}=-1)}{\sum_{i=1}^{n}\sum_{k=1}^{c}\mathbb{I}(G_{ik}=0)},$$

$$\mathrm{PLA} = \frac{\sum_{i=1}^{n}\sum_{k=1}^{c}\mathbb{I}(G_{ik}=0)\mathbb{I}(\widetilde{Y}_{ik}\neq -1)\mathbb{I}(\widetilde{Y}_{ik}=Y_{ik})}{\sum_{i=1}^{n}\sum_{k=1}^{c}\mathbb{I}(G_{ik}=0)\mathbb{I}(\widetilde{Y}_{ik}\neq -1)}.$$

Here, $\mathbb{I}(G_{ik} = 0)$ indicates that the k-th label of the i-th sample is unknown during training. $\widetilde{Y}_{ik} = -1$ denotes that CAPLA abstains from assigning a pseudo-label to this position, while $\widetilde{Y}_{ik} \neq -1$ indicates that a confident pseudo-label is assigned. It should be noted that the original label Y_{ik} at the masked positions is used only for diagnostic evaluation of pseudo-label quality and is never used for model training or parameter optimization.

During training, we further record the evolution of these three metrics, and the results are shown in Fig. 9. As observed, CAPLA maintains a high abstain rate at the early stage of training, indicating that the model adopts a conservative pseudo-labeling strategy when its predictions are still unstable. As training proceeds and the relaxation coefficients η_1 and η_0 gradually increase, the Total Fill Rate increases while the Abstain Rate decreases accordingly. This demonstrates that CAPLA progressively expands the coverage of confident pseudo-labels under the relaxation schedule.

Meanwhile, the Pseudo-label Accuracy remains consistently high throughout training, suggesting that the increased pseudo-label coverage does not introduce a large amount of inconsistent supervision. Since multi-label datasets are usually highly sparse, the generated pseudo-labels are dominated by negative pseudo-labels. Therefore, the overall pseudo-label accuracy also largely reflects the reliability of



(a) Corel5k (b) Pascal07

Fig. 8 Sensitivity of AP to contrastive temperature τ

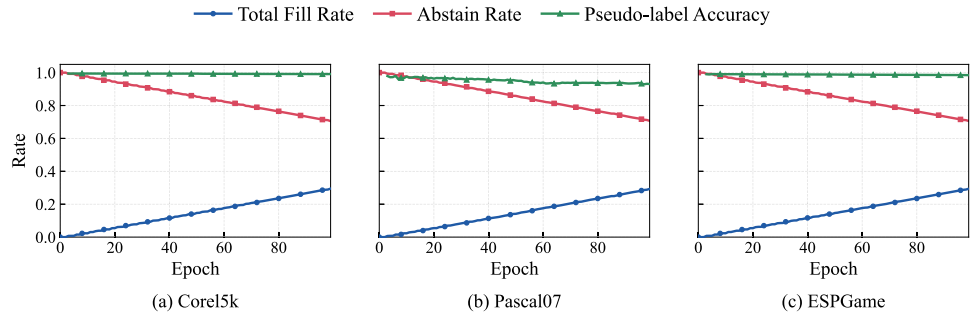
the generated high-confidence negative pseudo-labels under sparse label distributions.

4.6 Complexity and practical applicability

To further discuss the computational cost, scalability, and practical value of DGGI-Net, we analyze the model complexity and its applicability to real-world scenarios in this section. The notations are consistent with those defined above. Let $\bar{d} = \frac{1}{m}\sum_{v=1}^{m} d_v$ denote the average feature dimension of all views, K denote the number of nearest neighbors used in graph-guided imputation, and b denote the mini-batch size.

With the top-K sparse neighborhood strategy, DGGI-Net does not rely on all sample pairs for view completion, but instead retains only a limited number of reliable neighbors for each sample. Specifically, constructing and storing the sparse neighborhood graphs for all views requires approximately $O(mnK\bar{d})$ computational complexity and $O(mnK)$ storage complexity. The construction of the extended similarity graphs aggregates cross-view neighborhood information over the retained sparse edges, leading to a complexity of approximately $O(m^2nK)$. During graph-guided imputation, each missing instance aggregates features only from its top-K available neighbors, and therefore the imputation complexity is approximately $O(mnK\bar{d})$. The view-specific encoders introduce a complexity of $O(n\sum_{v=1}^{m} C_{\Psi^{(v)}})$, where $C_{\Psi^{(v)}}$ denotes the per-sample computational cost of the encoder for the v-th view. The dual-path fusion module and the classification heads are relatively lightweight, with complexities of $O(nmd_z)$ and $O(nd_zc)$, respectively. The contrastive consistency loss is computed within mini-batches, and its per-epoch complexity is approximately $O(m^2nbd_z)$. The neighborhood regularization loss is also computed based on local neighborhoods within mini-batches, and its per-epoch complexity is approximately $O(mnKd_z)$. CAPLA mainly involves category-adaptive threshold estimation and pseudo-label assignment. When sorting-based quantile estimation is used, its complexity is at most $O(cn\log n)$. Therefore, under the top-K sparse implementation, the overall per-epoch complexity of DGGI-Net can be summarized as $O(mnK\bar{d}+m^2nK+n\sum_{v=1}^{m} C_{\Psi^{(v)}}+m^2nbd_z+mnKd_z+nd_zc+cn\log n)$. Since $K \ll n$ in practical settings, the top-K sparse neighborhood strategy can effectively reduce the computational costs of graph storage, graph propagation, and missing-view completion. For larger-scale datasets, sparse graph storage and approximate nearest-neighbor search can be further adopted to improve scalability.

From the perspective of real-world applications, DGGI-Net is suitable for scenarios where view-level and label-level incompleteness coexist. Examples include medical image analysis, multimedia image annotation, remote sensing scene recognition, and autonomous driving perception. In medical image analysis, some patients may lack certain modalities such as CT or MRI due to acquisition cost, clinical protocols, or storage limitations, while diagnostic labels are often only partially annotated. In multimedia annotation and remote sensing scene recognition, different visual features or sensor views may be missing because of acquisition failure,



(a) Corel5k (b) Pascal07 (c) ESPGame

Fig. 9 Pseudo-label behavior of CAPLA during training on representative datasets, including Total Fill Rate, Abstain Rate, and Pseudo-label Accuracy

storage limitations, or heterogeneous collection conditions, and complete multi-label annotations are usually expensive to obtain. Similar issues may also arise in autonomous driving perception, where camera, LiDAR, or radar information may be partially unavailable under complex environments. Therefore, DGGI-Net can be trained offline using incomplete multi-view data and partial labels, and then deployed to make predictions under naturally occurring missing-view conditions.

4.7 Limitations and future work

This paper currently focuses on binary multi-label classification, where each label indicates whether a category is relevant or irrelevant to a sample. The graph-guided view completion module and the dual-path fusion strategy in DGGI-Net mainly operate at the feature representation level. Therefore, they are not strictly dependent on the binary label format and could potentially be transferred to other label spaces as framework-level components. However, the current BCE-based classification losses (Eqs. (19)–(20)) and the progressive relaxation, positive/negative thresholding, and pseudo-label assignment mechanism in CAPLA (Eqs. (13)–(16)) are specifically designed for binary labels, and thus cannot be directly applied to ordinal or continuous-valued label spaces without adaptation. For ordinal labels, the BCE losses would need to be replaced by ordinal regression objectives or cumulative-link based losses, while CAPLA could be extended to select high-confidence ordinal pseudo-labels based on calibrated level-wise probabilities or interval boundaries. For continuous-valued labels, regression losses such as MSE, MAE, or Huber loss would be more appropriate, and the current positive/negative thresholding strategy should be redesigned as a confidence- or uncertainty-aware pseudo-target selection mechanism. Therefore, DGGI-Net is potentially extensible at the framework level, but its supervision loss and pseudo-labeling module require task-specific adaptation according to the label format. We leave this direction for future work.

5 Conclusion

This paper proposes a Dynamic Graph-Guided Imputation Network (DGGI-Net) for multi-view multi-label learning with missing views and labels. The network is designed to jointly tackle view completion and supervision augmentation in a unified, end-to-end architecture. The dynamic graph-guided cross-view completion module adaptively constructs multi-level similarity graphs and leverages graph convolutions to aggregate neighborhood information, achieving high-fidelity reconstruction of missing views and producing representations that preserve local structures while maintaining cross-view consistency. Building upon the completed views, a dual-path multi-view fusion scheme separates original and imputed views into a high-quality fusion path and a low-quality fusion path, enabling the model to exploit complementary information from imputed views while limiting noise propagation. Furthermore, the Category-Adaptive Pseudo-Label Augmentation mechanism dynamically calibrates category-specific dual confidence thresholds from empirical label statistics and incorporates an abstention strategy to filter uncertain samples, thereby generating high-confidence pseudo-labels for unknown labels and enriching supervision under label sparsity. Experiments on five public datasets show that DGGI-Net consistently outperforms a series of strong baseline methods.

Author Contributions X.M. conceived the study, designed the method, conducted the experiments, analyzed the results, and wrote the original manuscript draft. Y.X. supervised the research, provided guidance on the methodology and manuscript organization, revised and polished the manuscript, and served as the corresponding author. Both authors reviewed and approved the final manuscript.

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Data Availability The datasets used in this study are publicly available benchmark datasets from existing sources. The processed data and experimental results generated during this study are available from the corresponding author upon reasonable request.

Declarations

Ethical Approval Not applicable.

Competing Interests The authors declare no competing interests.

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